

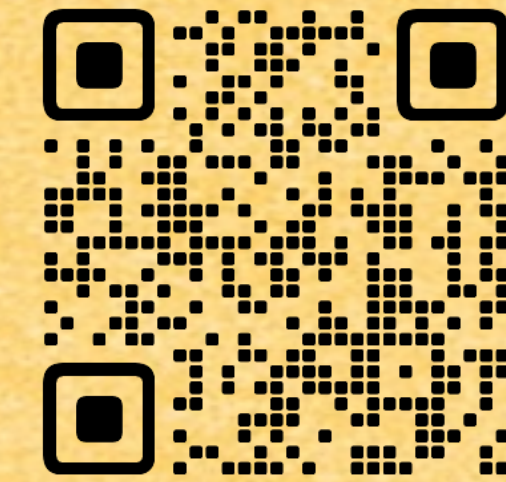
Beyond Utterance Length: Understanding Language Profiles in Natural Conversations of Autistic Youth



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Introduction

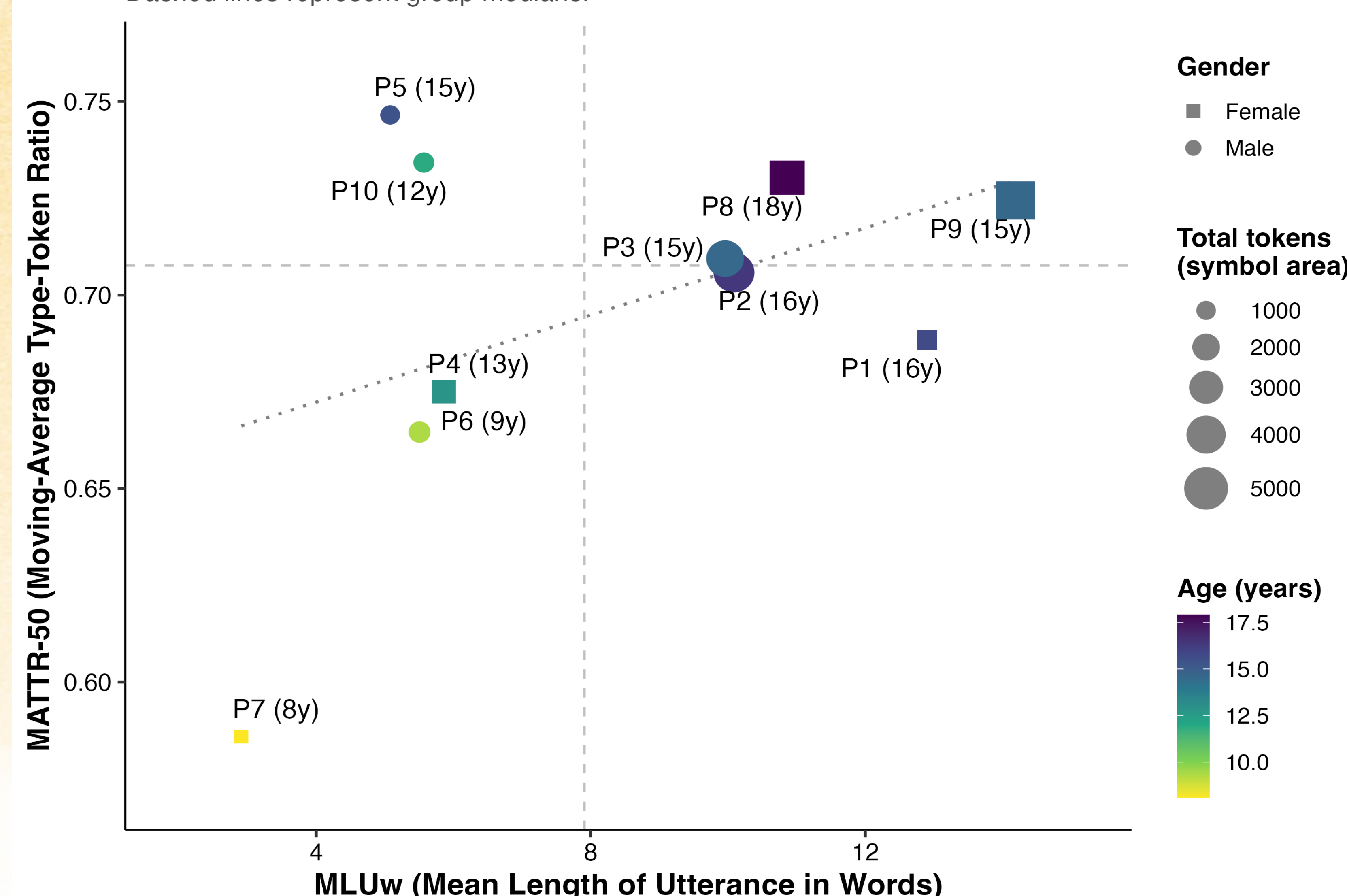
- **Language** abilities vary widely among autistic youth, and a single measure may obscure uneven profiles across language dimensions.
- Spontaneous **speech** metrics such as utterance length (MLUw) and lexical diversity (TTR/MATTR) offer non-invasive proxies for cognitive-linguistic capacity.
- Focusing on **natural conversations**, this study examined how utterance length and lexical diversity coordinate across age, revealing diverse profiles of **individual** language patterns.
- We hypothesized that longer utterances would not consistently coincide with greater lexical diversity, revealing **variation** that either measure alone would miss.

Preliminary Results

- Age was positively correlated with utterance length (MLUw: $r = 0.77$, $p = .010$) and lexical diversity (MATTR: $r = 0.76$, $p = .011$). Age was also significantly associated with the language measures examined jointly (MANOVA: Wilks' $\lambda = 0.15$, $F(3, 5) = 9.74$, $p = .016$).
- Longer utterances did not consistently contain more diverse vocabulary, highlighting individual differences across language dimensions.

Figure 1. Language Profiles across Participants (n = 10)

Dashed lines represent group medians.



Conclusions

- Although older participants generally produced longer utterances and more lexically diverse speech, MLUw and MATTR-50 were weakly correlated, indicating that these dimensions do not necessarily vary together and that a single measure may obscure uneven individual language profiles.
- Smartwatch recordings combined with semi-automated analysis can reveal complementary dimensions of everyday language beyond how much someone speaks.
- The small sample and cross-sectional age comparisons limit conclusions about developmental change. These exploratory language profiles require validation in larger samples.
- Next, larger longitudinal studies will test profile stability across contexts and over time, examine repetitive wording, syntactic elaboration, and silent and filled pauses, and validate MATTR against clinician-rated change.

References

- Boo, C. (2021). Structural and Pragmatic Language Abilities of Children with ASD ADHD and Comorbid Symptoms of Both During Conversational Discourse in a Virtual Reality Task. University of Connecticut.
- Covington, M. A., & McFall, J. D. (2010). Cutting the Gordian knot: The moving-average type-token ratio (MATTR). *Journal of quantitative linguistics*, 17(2), 94-100.
- Kalagotla, S. G., Wang, F., Datla, K., Ferguson, B., & Beversdorf, D. Q. (2026). Toward a Smartwatch-Based System for Automated Social Interaction Monitoring in Autism. *IEEE Access*.
- Kyle, K., Crossley, S. A., & Jarvis, S. (2021). Assessing the validity of lexical diversity indices using direct judgements. *Language Assessment Quarterly*, 18(2), 154-170.
- Maes, P., La Valle, C., & Tager-Flusberg, H. (2024). Frequency and characteristics of echoes and self-repetitions in minimally verbal and verbally fluent autistic individuals. *Autism & Developmental Language Impairments*, 9, 23969415241262207.
- Tuller, L., Silleresi, S., & Prévost, P. (2025). Autism and language modularity. *Footprints of Phrase Structure*, 288, 36-54.

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Methods

